

SE Lab - 3

Component Modelling & Architectural Pattern

Problem Statement #08

Digital Campus Library Reservation Gateway

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Written Justification

Architecture Selection

We chose a **Layered Architecture** for the Digital Campus Library Reservation Gateway.

Architectural Choice

The system is organized into horizontal layers with distinct responsibilities: a Presentation Layer (UI), a Business/Application Layer (Catalog, Hold, Queue, Fine, and Auth Managers), and a Data Layer (Database).

Reason 1 - Separation of Responsibilities

The system handles distinct functional operations: ISBN catalog searching (FR-001), hold placement (FR-002), FIFO queue logic (FR-003), and fine tracking (FR-005). A layered structure isolates these business rules from the user interface and database schema. This decoupling ensures that updates to one component (e.g.,

modifying notification logic) do not disrupt the presentation layer or other business modules.

Reason 2 - Consistent Centralized Data

The reservation gateway requires continuous, shared access to interrelated records (catalog items, member profiles, active holds). Funneling all business components through a common data layer establishes a single source of truth, preventing data anomalies and sync errors when multiple users interact with the library catalog simultaneously.

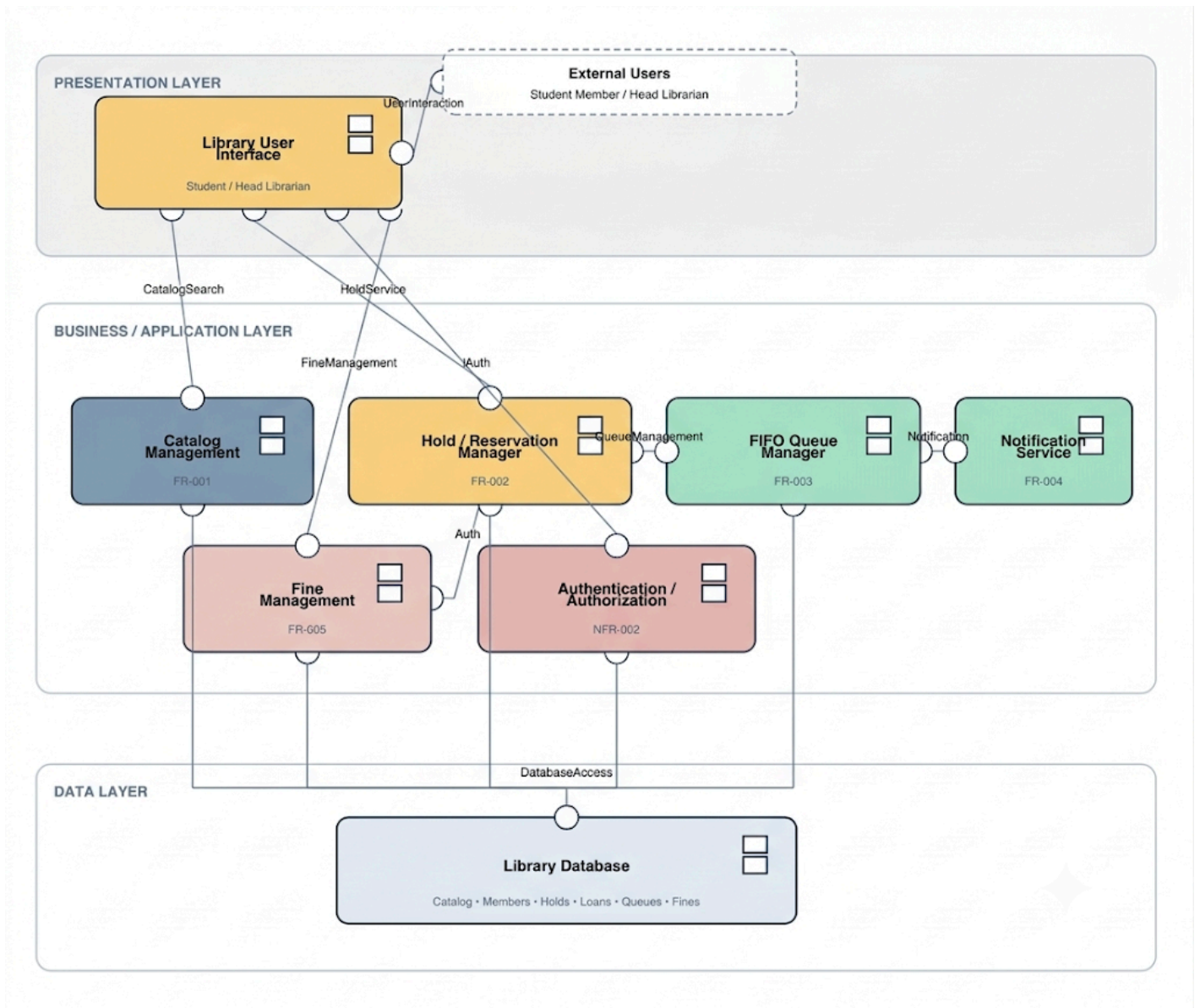
Security Advantage

Authorization and access control (NFR-002) are strictly enforced at the Business Layer via an Authentication component before any sensitive queries reach the Data Layer. This guarantees that privileged librarian-only functions, like waiving fines or editing catalog entries are fundamentally inaccessible to standard Student Members, minimizing the system's attack surface.

Performance Benefit

By decoupling the database interface from the UI, the Catalog Management component can execute optimized, paginated queries natively at the data layer. This ensures only the requested subset of data is retrieved rather than loading the entire database into memory, fulfilling the requirement to return search results for 50,000 titles in under 150 milliseconds.

Component Diagram



Legend



Provided interface (ball)



Required interface (socket)



Assembly connector

Maps to Lab 1: FR-001 search | FR-002 holds | FR-003 FIFO queue | FR-004 notification | FR-005 fines | NFR-002 access control

8 interfaces